

Literature on Forecasting with Graph Neural Networks

1 Temporal Graph Learning

- Expressivity of Representation Learning on Continuous-Time Dynamic Graphs: An Information-Flow Centric Review (<https://arxiv.org/pdf/2412.03783>)
- Temporal Graph Benchmark for Machine Learning on Temporal Graphs (<https://arxiv.org/pdf/2307.01026>)
- Learning production functions for supply chains with graph neural networks (<https://arxiv.org/abs/2407.18772>)
- Graph Deep Learning for Time Series Forecasting (Andrea Cini) (<https://arxiv.org/pdf/2310.15978>)
- Graph Neural Networks for temporal graphs: State of the art, open challenges, and opportunities (<https://arxiv.org/pdf/2302.01018>)
- Deep Learning for Time Series Forecasting: Tutorial and Literature Survey (<https://arxiv.org/pdf/2004.10240>)
- Temporal graph kernels for classifying dissemination processes (<https://arxiv.org/abs/1911.05496>)
- Temporal Graph Reading (<https://shenyanghuang.github.io/rg.html>)

2 Graph Neural Networks

- Representation Learning on Graphs: Methods and Applications (<https://arxiv.org/pdf/1709.05584>)
- How Powerful are Graph Neural Networks (<https://arxiv.org/pdf/1810.00826>)
- GNNExplainer: Generating Explanations for Graph Neural Networks (<https://arxiv.org/abs/1903.03894>)
- On the expressive power of graph neural networks: beyond the Weisfeiler-Leman test (https://usiena-air.unisi.it/bitstream/11365/1265054/3/phd_unisi_103967.pdf)
- On the Expressive Power of Geometric Graph Neural Networks (<https://arxiv.org/pdf/2301.09308>)
- Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges (Michael Bronstein) (<https://arxiv.org/abs/2104.13478>)

- Graph Foundation Models: Concepts, Opportunities and Challenges (<https://arxiv.org/pdf/2310.11829>)
- How Expressive are Knowledge Graph Foundation Models? (<https://arxiv.org/pdf/2502.13339?>)
- Combinatorial optimization and reasoning with graph neural networks (<https://arxiv.org/pdf/2102.09544>)

3 Oversquashing and Oversmoothing

- On the Trade-off between Over-smoothing and Over-squashing in Deep Graph Neural Networks (<https://arxiv.org/pdf/2212.02374>)
- A Survey on Oversmoothing in Graph Neural Networks. (<https://arxiv.org/pdf/2303.10993>)
- Unifying over-smoothing and over-squashing in graph neural networks: A physics informed approach and beyond (<https://arxiv.org/pdf/2309.02769>)
- Spectral Graph Pruning Against Over-Squashing and Over-Smoothing (<https://arxiv.org/pdf/2404.04612>)

4 Networks

- Position: Graph Learning Will Lose Relevance Due To Poor Benchmarks (<https://arxiv.org/pdf/2502.14546>)
- Motifs in Temporal Networks (<https://dl.acm.org/doi/pdf/10.1145/3018661.3018731>)
- Echoes of the hidden: Uncovering coordination beyond network structure (<https://arxiv.org/pdf/2504.02757>)

5 Generalization of Graph Neural Networks

- Optimisation & Generalisation in Networks of Neurons (<https://arxiv.org/abs/2210.10101>)
- Survey on Generalization Theory for Graph Neural Networks (Christopher Morris) (<https://arxiv.org/pdf/2503.15650>)
- Generalization of Graph Neural Networks through the Lens of Homomorphism (<https://arxiv.org/pdf/2403.06079>)
- Generalization of Graph Neural Networks is Robust to Model Mismatch (<https://arxiv.org/html/2408.13878v2>)
- Out-Of-Distribution Generalization on Graphs: A Survey (<https://arxiv.org/pdf/2202.07987>)
- Towards Understanding Generalization of Graph Neural Networks (<https://proceedings.mlr.press/v202/tang23f/tang23f.pdf>)